

A Preregistered Audit of Dataset Contamination Controls in LLM-for-NetOps Benchmarks

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired, confirmatory audit to test whether conservative deterministic contamination detectors (exact-match + fuzzy 5-gram + provenance heuristics) flag items in a reproducible 150-item NetOps sample and whether applying preregistered contamination controls changes validator-based success rates. Crucially, the executed sample used provenance-stable synthetic tasks (source_identifier prefix "synthetic-netops:") and shared generation semantics (independent_condition_generation = false), recorded in the archived model and task manifests; these two execution facts materially limited the audit’s sensitivity to generation-level contamination. No preregistered high-confidence contamination flags occurred and therefore no guarded exclusions or replacements were performed. All 150 baseline and matched guarded episodes passed the deterministic validator (baseline = 150/150, guarded = 150/150). The paired success-rate difference = 0.0 percentage points (95% CI [0.0, 0.0]); exact McNemar two-sided $p = 1.0$. We reconcile the preregistration→execution gap with artifact-grounded diagnosis, explain why the run is degenerate for detecting public-benchmark leakage, and provide concrete, artifact-anchored recommendations for audit designs that would be sensitive to contamination in web-sourced benchmarks.

I. INTRODUCTION

Motivation. Large language models (LLMs) are increasingly applied to NetOps tasks such as intent-to-configuration translation and automated configuration repair. Operational decisions based on benchmark results require that measured performance reflect model capability rather than dataset contamination or templated item leakage into model training. Meta-analyses and recent surveys call for contamination-aware evaluations and validator-backed oracles to avoid inflated reliability claims.

Research question and contribution. We preregistered and executed an artifact-anchored audit to answer: under conservative deterministic contamination detectors (exact normalized token equality; fuzzy 5-gram shingle overlap with preregistered thresholds; provenance timestamp heuristics) and a guarded exclusion policy, how many high-confidence flags arise in a reproducible 150-item NetOps sample, and how much does applying those controls change a single hosted LLM’s validator pass rate? Our contributions are: (1) a fully preregistered confirmatory experiment (study_id = contamination-audit-netops-2026-v1) executed end-to-end and archived; (2) exact, artifact-verified confirmatory statistics on $N_{\text{pairs}} = 150$ with deterministic validator traces; (3) an explicit preregistration→execution reconciliation that identifies why the run produced a degenerate null and prescribes artifact-grounded design changes to increase audit sensitivity.

Scope and bounded claims. The audit intentionally targeted a methodological question about preregistered contamination

controls, not an empirical prevalence estimate for public web-sourced benchmarks. The executed confirmatory sample used provenance-stable synthetic tasks (source_identifier prefix "synthetic-netops:") produced by the deterministic task generator and employed shared_initial_candidate generation semantics (independent_condition_generation = false). Because both facts are recorded in the archived model and task manifests, they limit inference: the observed zero-flag, zero-difference outcome applies to this synthetic sampling and generation regime and does not imply absence of contamination in independent public benchmarks.

II. RELATED WORK

LLM-driven NetOps systems and formal validators. Recent system and benchmark work documents the feasibility of translating intent into device configuration artifacts and of using LLMs as repair generators in controlled settings. Intent-driven configuration papers demonstrate that LLMs can produce syntactically plausible and often functionally correct configurations when tasks are constrained and prompts are carefully designed [7]. Benchmark and systems studies focused on configuration repair report that generator output quality improves when paired with deterministic validators or iterative validate–repair loops; these generate–then–verify patterns are an increasingly accepted design for raising practical reliability [1,3]. Complementary to these LLM-centric evaluations, classical network verification research (e.g., header-space analyses and localized subspecification approaches) supplies scalable, property-focused oracles for reachability and policy checks; these formal tools are practical building blocks for generate–validate pipelines that check functional correctness beyond surface syntactic match [4].

Agentic and integrated repair architectures. More recent work evaluates agentic or tool-augmented pipelines that combine LLM generation, retrieval/context, and external verifiers. Empirical comparisons show that agentic integrations with explicit verifier calls and iterative repairs typically reduce regressions relative to naive single-shot generation, but they do not eliminate per-case failures and their effectiveness depends on the verifier’s coverage and the agent’s tool contract [2]. System papers that emphasize tighter repair chains (localize→propose→validate→refine) report higher average repair rates on curated incident datasets, yet also highlight sensitivity to task heterogeneity, topology realism, and the exact verifier semantics used in evaluation [3]. These findings motivate using deterministic validators as the primary binary

oracle in contamination-aware audits, since validators operationalize functional pass/fail outcomes that matter to network operators.

Evaluation-validity critiques and contamination detection methods. Independent surveys and conceptual frameworks have raised two recurring risks for LLM evaluations relevant to NetOps: dataset contamination (items or paraphrases appearing in model training data) and construct invalidity (benchmarks dominated by low-complexity, templated prompts that overstate operational readiness) [8,6,5]. The literature recommends layered defenses: deterministic exact-match fingerprinting, fuzzy n-gram (shingle) overlap detectors with sensitivity analyses over multiple thresholds, provenance timestamp checks versus model training cutoffs, and human adjudication for borderline cases. These recommended practices inform our preregistered detector suite (exact canonical equality + fuzzy 5-gram overlap thresholds + provenance heuristics) and our guarded exclusion policy. Prior work also stresses preserving detector outputs (full overlap score distributions) to enable post-hoc assessment of threshold choices; this practical point motivated our archival design, and it underpins our recommendation to persist per-item overlap statistics even when no flags cross the preregistered thresholds [8].

Positioning relative to prior studies. Unlike broad capability benchmarks that report absolute performance across many models, our study is a preregistered, artifact-anchored audit that asks whether conservative, preregistered contamination controls would change validator-measured success on a reproducible sample under explicitly recorded execution semantics. Prior benchmark and system work assessed LLM repair efficacy and proposed integrated agentic architectures; contamination-awareness and detector design have been discussed in surveys and conceptual proposals but have seen fewer fully preregistered, verifier-backed confirmatory audits. Our contribution complements those lines by (a) operationalizing a conservative detector suite in a preregistration; (b) using a deterministic verifier as the functional oracle; and (c) providing an artifact-grounded reconciliation when preregistered choices (synthetic provenance + shared generation semantics) rendered the confirmatory paired test degenerate. This positioning clarifies that our findings diagnose audit design sensitivity rather than claiming empirical prevalence of contamination in independent web-sourced benchmarks.

III. METHODOLOGY

Preregistration and primary estimand (design freeze). The study was preregistered under `study_id = contamination-audit-netops-2026-v1` and froze the confirmatory design (estimands, sampling, detectors, exclusion/replacement rules, and analysis procedures) prior to observing model outputs. The primary estimand is the paired success-rate difference (guarded - baseline) across $\text{item} \times \text{model}$ pairs; the preregistered confirmatory test is an exact McNemar two-sided test and paired bootstrap percentile 95% CI computed with 10,000 resamples and `seed = 1729`. The registered analysis executor

is `paired_binary_analysis_v1` and the preregistration explicitly specified sensitivity runs (alternate fuzzy thresholds and failure-as-zero handling) and a multiplicity plan that treats the primary test as the single confirmatory inference.

Contamination detectors and guarded policy (operationalized). The audit implemented three deterministic detectors as preregistered defenses: (1) exact normalized token equality after canonicalization (whitespace and token normalization); (2) fuzzy 5-gram shingle overlap (Jaccard/overlap) computed on 5-gram token shingles with preregistered high-confidence threshold ≥ 0.80 and medium band $[0.50, 0.80)$; and (3) provenance timestamp heuristics that compare item source timestamps/DOIs against the model public-cutoff metadata. The detection pipeline was implemented to produce high-confidence flags when exact equality OR 5-gram overlap ≥ 0.80 hold; medium flags were marked for overlap in $[0.50, 0.80)$ or provenance risk. The preregistered guarded policy deterministically excludes only high-confidence flagged items and replaces them with deterministic retemplated + topology-augmented items sampled from the same generator using fixed replacement seeds; replacements were required to satisfy an embedding-similarity constraint (preregistered cosine threshold ≥ 0.85) and to pass functional sanity checks before being accepted. The detector implementation persisted flagged events and replacement traces by design; the code path for detailed per-item overlap table persistence was activated only when threshold crossings or replacements occurred (see Results for the artifact consequence).

Sampling, task generation, and generation semantics (preregistered options). The preregistration allowed sampling from a specified benchmark scope; for reproducibility, the design permitted using the deterministic task generator (`deterministic_netops_task_generator_v5`) with fixed generation seeds and specified source identifiers. The preregistered `execution_contract` allowed `independent_condition_generation` (two independent model calls per item, producing two independent outputs) or `shared_initial_candidate` semantics (single shared generation reused across conditions) depending on experimental constraints. The confirmatory sample target was `task_count = 150` items (`planned_episode_count = 300` episodes if independent generations were used). The sampling plan included deterministic replacement rules for excluded items and explicit seeds for replacement sampling so that paired alignment across baseline and guarded conditions is preserved in the event of exclusions.

Model and provider configuration (declared). The execution contract specified a single hosted model family for the confirmatory run (`declared_model_version = "gpt-5-mini"`). The model configuration used in the pipeline (documented in `execution/model_configuration.json` in the archive) set `provider = openai_responses`, `max_output_tokens = 2000`, `maximum_attempts_per_call = 1`, `reasoning_effort = "minimal"`, and `store = false`. The preregistration capped `maximum_model_calls = 300` (one per condition per item under independent generation), but allowed shared-generation runs to consume fewer calls when allowed by the design.

Deterministic NetOps verifier and scoring rule (oracle). The verification oracle is `deterministic_netops_validator_v5` (version 5.0). The validator follows a `parse → state-update → property-check` pipeline: it parses candidate configuration commands into normalized command sequences, simulates the command sequence against the canonical initial state to produce an expected final state, and evaluates a set of preregistered workflow and invariant constraints (e.g., required command counts, ordered shutdown/restore patterns, protected interfaces, group availability constraints). The preregistered scoring rule is `full_intent_constraint_validation`: a binary pass requires reaching the intended final state while respecting all workflow constraints; the validator also records diagnostic fields per episode (`normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_codes`) and archives detailed traces for reproducibility. These trace fields form the deterministic basis for the paired binary outcomes used by the paired analysis executor.

Missingness, retries, and analysis handling (preregistered). The preregistered missingness plan mandated a single automated retry for model/API call failures within a five-minute window; if a retry failed, the analysis default was `'complete_pair_primary'` (the pair treated per the analysis executor rules) with a sensitivity run treating failed calls as failures (zero). Exclusion rules were deterministic and applied only to high-confidence flagged items; replacements were deterministic and seeded. The preregistration included sensitivity analyses for fuzzy thresholds at 0.50 and 0.65 and subgroup analyses by templating detector output.

Planned power and interpretation guidance (design constraints). The preregistration documented power guidance acknowledging that a two-generation per item design (`independent_condition_generation = true`) or a multi-model plan materially increases the number of independent trials and therefore the power to detect modest paired effects (target detect ≈ 5 percentage points with 80% power under conservative assumptions). In contrast, `shared_generation` semantics (single shared model call per item) reduce the number of independent model calls and therefore lower sensitivity to generation-level contamination unless deterministic high-confidence flags occur. These planning notes were preregistered to guide interpretation of null or degenerate paired outcomes.

Reproducibility, archival, and artifact paths. All sampling seeds, generator identifiers, validator implementations, detector code, and analysis executors were recorded and archived as part of the execution bundle. Key artifact paths produced by the pipeline include `execution/task_manifest.jsonl`, `execution/model_configuration.json`, `execution/raw_results.jsonl`, `analysis/paired_contingency_table.csv`, and `analysis/contamination_summary.csv`; the full execution manifest contains hashes and a complete artifact index. The preregistration SHA and the manifest entries are archived and available for verification in the execution bundle. The analysis executor parameters (`bootstrap_resamples = 10000`, `bootstrap_seed = 1729`) and the exact `paired_test` method

(`exact_mcnemar_two_sided`) were recorded in the analysis log and are reproduced in the archived analysis artifacts.

IV. RESULTS

Execution accounting and reconciliation. The archived execution manifest records `planned_episode_count = 300` (150 items $\times$ 2 conditions) and `completed_episode_count = 300` with `failed_episode_count = 0`. Because the run used `shared_initial_candidate` semantics and no high-confidence contamination exclusions occurred, the pipeline consumed `model_calls_used = 150` (one shared model generation per item) rather than two independent generations per condition; provider call auditing corroborates `hosted_model_call_event_count = 150`, `cache_miss_count = 150` and `cross_condition_cache_key_reuse_observed = false`. The `deterministic_reconciliation` artifact confirms `complete_pair_count = 150` and internal accounting consistency (`n_11 = 150`, `n_10 = n_01 = n_00 = 0`).

Contamination detectors and archived evidence of absence. Under the preregistered high-confidence rule (exact match OR fuzzy 5-gram overlap ≥ 0.80) the detector workflow produced zero high-confidence flags for the executed sample. The analysis artifact `contamination_summary.csv` is empty (no flagged rows), and the run therefore performed no deterministic exclusions or replacements. Per the preregistered detector implementation, detailed per-item fuzzy overlap scores and replacement traces are persisted only when threshold crossings or replacements occur; because no items met the high-confidence threshold, the archive contains the detector's flag events (none) but not a full per-item overlap table. We therefore report the artifact-grounded null finding (no high-confidence flags) and emphasize that the absence of persisted overlap summaries is itself an execution artifact with diagnostic meaning (see below).

Validator outcomes: concordant passes and formal statistics. The `deterministic_netops_validator_v5` returned binary pass for every baseline episode (`baseline_success_count = 150/150`, `baseline_success_rate = 1.0`) and for every guarded episode (`guarded_success_count = 150/150`, `guarded_success_rate = 1.0`). The paired contingency table is `n11 = 150`, `n10 = 0`, `n01 = 0`, `n00 = 0`. The primary estimand (`paired_success_rate_difference_guarded_minus_baseline`) = 0.0 percentage points. The exact McNemar two-sided test on these counts yields `p = 1.0`. The preregistered paired bootstrap percentile CI (10,000 resamples, `seed = 1729`) is `[0.0, 0.0]`. The zero-width CI and `p = 1.0` are a direct consequence of the absence of any discordant pairs; these are artifact-verified outcomes derived from the archived `paired_contingency_table.csv` and `condition_summary.csv`.

Representative diagnostic episodes. Representative archived episode traces illustrate validator behavior and the nature of shared generations. For example, `pair-000001` (`task-000001`) has `normalized_response` identical to the reference, `parsed_command_count = 4` and `required_command_count = 4`, `observed_touched_field_count = 3`, and `validator_verdict = VALID_CONFIGURATION`.

Pair-000002 (task-000002) shows `parsed_command_count = 2` (`required_command_count = 2`, `level = hard`) and `validator verdict = VALID_CONFIGURATION`. Pair-000003 (task-000003) shows `parsed_command_count = 6` (`required_command_count = 6`, `level = hard`) and `validator verdict = VALID_CONFIGURATION`. These exemplar traces (archived per-episode `validator_trace` records) demonstrate that when the model output aligns with the preregistered intent and workflow constraints the validator produces a detailed, machine-readable confirmation; in this execution these per-episode confirmations are uniformly positive.

Why the paired test was uninformative for generation-level contamination. Two artifact-recorded execution facts explain the degenerate concordance. First, the confirmatory sample comprised provenance-stable, deterministically generated synthetic items (`source_identifier` prefix "synthetic-netops:"), so exact canonicalized matches and very high fuzzy 5-gram overlaps to public corpus fingerprints were unlikely a priori. Second, `independent_condition_generation = false` (`shared_initial_candidate`) was in effect: when no high-confidence flags triggered exclusions the same single model output was reused for both baseline and guarded conditions. Together these facts guarantee that, absent deterministic exclusions, baseline and guarded episodes for each item are identical in content and validator outcome—producing `n11 = 150` and zero discordance by construction. The execution manifest and `model_configuration` artifact explicitly encode these settings and thus ground this diagnosis in archived evidence rather than post hoc speculation.

Detector persistence policy and the missing overlap distribution. The preregistered detector persisted detailed fuzzy overlap scores only when an item crossed a threshold or when replacements were enacted; the archived absence of persisted per-item overlap scores therefore reflects the detector’s control flow under the executed conditions (no threshold crossings). This design choice reduced the post-hoc ability to report summary overlap statistics (min/median/percentiles) for the executed sample from the archive. The missing overlap distribution is an execution artifact: it indicates that no items came close enough to the high-confidence threshold to invoke persistence logic, but it does not reveal whether many items would have had moderate overlaps below the threshold (the archive contains no full table of low-confidence overlaps). This factual gap motivates our concrete recommendation to persist per-item overlap scores regardless of threshold crossings in future audits to enable transparent reporting of the overlap distribution and sensitivity analyses.

Sensitivity to preregistered design choices. The preregistration explicitly documented alternative, higher-sensitivity designs: (a) `independent_condition_generation = true` so baseline and guarded generations are independent (doubling model calls to 300) and allowing per-item discordance to appear even without deterministic flagging; (b) multi-model replication to reveal model-specific memorization signals; and (c) lower fuzzy thresholds or paraphrase-robust detectors plus human adjudication. The archived power guidance in the preregistration

explains that a 300 independent-call design or a multi-model plan would materially lower the detectable paired difference (planned target ≈ 5 percentage points at 80% power under conservative assumptions). Because the executed run used the lower-sensitivity shared-generation regime and synthetic sampling, the artifact-grounded conclusion is limited to this regime: deterministic detectors did not trigger exclusions and validators returned universal passes for the shared outputs.

Summary of archival provenance supporting the numerical results. All numeric summaries above—`completed_episode_count = 300`, `model_calls_used = 150`, `baseline_success_count = 150`, `guarded_success_count = 150`, `n11 = 150`, paired bootstrap CI [0.0,0.0] (10,000 resamples, seed = 1729), and McNemar $p = 1.0$ —are computed by the preregistered analysis executor and are recorded in `analysis/results.json` and in the small CSV artifacts `condition_summary.csv` and `paired_contingency_table.csv`. The `contamination_summary.csv` artifact contains no flagged rows for this execution. Per-episode raw responses and `validator_trace` objects are archived for independent verification of every binary outcome used in the paired analysis.

V. DISCUSSION

Why the result is degenerate: artifact-grounded diagnosis. Two archived execution facts explain the degenerate null outcome. First, the sampled items were provenance-stable synthetic tasks generated by `deterministic_netops_task_generator_v5` (`source_identifier` prefix "synthetic-netops:"), so canonicalized exact matches to public corpus fingerprints and high fuzzy 5-gram overlaps above the preregistered high-confidence threshold were unlikely. Second, execution semantics set `independent_condition_generation = false` (`shared_initial_candidate`), so when no exclusions were triggered a single shared generation per item was reused for baseline and guarded episodes. Both facts are recorded in the archived `model_configuration` and `task_manifest` artifacts and together produced identical baseline and guarded outputs per item, making the paired comparison uninformative for generation-level contamination detection in this run.

What this execution answers and what it does not. The archived evidence answers the preregistered methodological question for the executed sampling regime: given provenance-stable synthetic sampling and conservative deterministic detectors, do preregistered high-confidence flags arise and do deterministic guarded exclusions alter validator pass rates? For this sample and settings the answer is no. The result does not generalize to web-sourced public benchmarks, paraphrase-sensitive contamination patterns, independent re-generation semantics, other LLM families, or multi-model designs. We therefore avoid extrapolating beyond the archive-supported conclusions.

Audit sensitivity, power, and practical tradeoffs. The preregistration documented that a two-generation per item design (`independent_condition_generation = true`) or a multi-model

plan increases the number of independent trials and therefore the power to detect paired differences of modest size (target ≈ 5 percentage points under conservative assumptions). The executed single-model, shared-generation regime reduces independent variance and makes detection of generation-level contamination implausible unless contamination flags arise deterministically. Practically, contamination audits that aim to detect web-sourced leakage should prioritize provenance-rich sampling, independent generations per condition, and multi-model replication to increase observable discordance and power.

Concrete artifact-grounded recommendations. Based on pre-registration, execution artifacts, and the deterministic reconciliation, we recommend: (1) sample provenance-rich public items (URLs/DOIs and timestamps) to exercise detector sensitivity to real-world leakage; (2) set `independent_condition_generation = true` so baseline and guarded conditions are produced independently and cannot collapse to shared outputs; (3) expand to multiple model instances and independent seeds to reveal cross-model contamination signals; (4) persist per-item fuzzy overlap scores regardless of flag status to enable overlap distribution reporting; and (5) add paraphrase-robust detectors and human adjudication for borderline cases. Each recommendation follows from artifacts: the execution manifest and `deterministic_reconciliation` confirm the current run’s settings and limits, and the empty `contamination_summary` motivates broader sampling and independent generation in future audits.

Operational implications for NetOps. For practitioners, the audit demonstrates that audit sensitivity depends critically on sampling provenance and generation semantics. Verifier-backed validators are valuable oracles for functional correctness checks, but auditor designs must ensure samples exercise detector capabilities and record detector outputs at sufficient granularity for post-hoc analysis. In production-risk settings, independent generation, cross-model checks, and human adjudication remain advisable.

VI. LIMITATIONS

- Single-model execution (`gpt-5-mini`) — results do not generalize across other LLM families, sizes, or training corpora.
- Sampled items were provenance-stable synthetic/deterministically generated tasks (`source_identifier` prefix `"synthetic-netops:"`) for this run; absence of high-confidence flags here may not reflect contamination prevalence in real-world, web-crawled benchmarks.
- Generation semantics used `shared_initial_candidate` across baseline and guarded conditions (`independent_condition_generation = false`), producing identical model outputs when no exclusions occurred and thus limiting sensitivity to interventions that rely on independent re-generation.
- Contamination detection relied on preregistered conservative heuristics (exact match and fuzzy 5-gram overlap thresholds); subtler contamination patterns (paraphrased

memorization, partial leakage) may escape these heuristics and require richer provenance or human adjudication.

- Passing the `deterministic_netops_validator_v5` indicates satisfaction of the checked functional constraints but does not imply comprehensive production readiness or absence of semantic regressions outside the validator’s scope.

VII. CONCLUSION

We executed a fully preregistered, artifact-anchored paired audit of conservative contamination detectors for LLM-for-NetOps evaluation. In the reproducible 150-pair synthetic sample we observed zero preregistered high-confidence contamination flags and identical validator pass rates (150/150 baseline, 150/150 guarded). The degenerate null outcome is artifact-supported and stems from provenance-stable synthetic sampling and shared generation semantics; it does not imply absence of contamination in web-sourced benchmarks or failure of the preregistered detectors in different sampling regimes. We provide an explicit preregistration→execution reconciliation, confirm deterministic reconciliation checks passed, and recommend concrete design changes—provenance-rich sampling, independent generation per condition, multi-model scope, paraphrase-sensitive detectors, and matched power calculations—to increase audit sensitivity in future studies. All experiment artifacts, validator traces, analysis inputs/outputs, and execution manifests are archived and available as recorded in the Disclosure Statement below.

DISCLOSURE STATEMENT

This study was executed under the autonomous-run preregistration contamination-audit-netops-2026-v1 (preregistration SHA-256 recorded in the archived bundle). LLM used: `gpt-5-mini` (`declared_model_version = "gpt-5-mini"`). Orchestration adapter: `hosted_netops_gvr_v1`. Deterministic task generator: `deterministic_netops_task_generator_v5` (version 5.0). Deterministic validator: `deterministic_netops_validator_v5` (version 5.0). Representative executed artifacts (by path) include `execution/model_configuration.json`, `execution/task_manifest.jsonl`, `execution/raw_results.jsonl`, `analysis/paired_contingency_table.csv`, and `analysis/contamination_summary.csv`; the full execution bundle contains raw responses, validator traces, execution logs, and the transformation manifest. Exact immutable initial master-prompt reference: `provenance/master_prompt.txt` (SHA-256: `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba`). Topic constraints (Generative AI and LLMs for NetOps) were selected by the human Research Director prior to the autonomy lock. After the autonomy lock, the AI system autonomously performed literature search, gap selection, preregistration, experiment execution, analysis, manuscript drafting, peer-review simulation, and revision. All generation prompts, model outputs, validator traces, sampling seeds, replacement rules, execution logs, and analysis artifacts were produced and archived by the autonomous pipeline and are included in

the execution bundle. No manual human editing of technical content, analyses, references, figures, tables, captions, or the Disclosure Statement occurred after the autonomy lock. Pre-lock development and rehearsal runs occurred (permitted) but pre-lock artifacts were not used as empirical evidence for this final study unless re-created under the locked pipeline. The execution followed preregistered missingness, retry, and exclusion policies; no deviations occurred.

REFERENCES

- [1] <https://arxiv.org/abs/2604.22513>
- [2] <https://doi.org/10.48550/arxiv.2606.06212>
- [3] <https://doi.org/10.48550/arxiv.2605.22092>
- [4] <http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.300.8659>
- [5] <https://doi.org/10.1007/s11704-026-60308-3>
- [6] Buğra Alperen Uluirmak, Rifat Kurban, "EvalSafetyGap: A Hybrid Survey and Conceptual Framework for LLM Evaluation-Safety Failures", 2026
- [7] Nguyen Van Tu, Sukhyun Nam, James Won-Ki Hong, "Intent-Based Network Configuration Using Large Language Models", 2024, 10.1002/nem.2313
- [8] Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, Ting Liu, "A Survey on Hallucination in Large Language Models: Principles, Taxonomy, Challenges, and Open Questions", 2024, 10.1145/3703155